



Overview

This document references the local control protocol using the device's TCP port. This protocol supports the following connection method:

- Local WiFi network communication using HTTP/IP GET requests

Before using the local control communication, set up the controller and blinds using our “Neo Smart Blinds” app found on [Google Play](#), [Itunes](#), or the web [Chrome](#) version. Follow the in-app instructions for adding the controller(s) and blind(s).

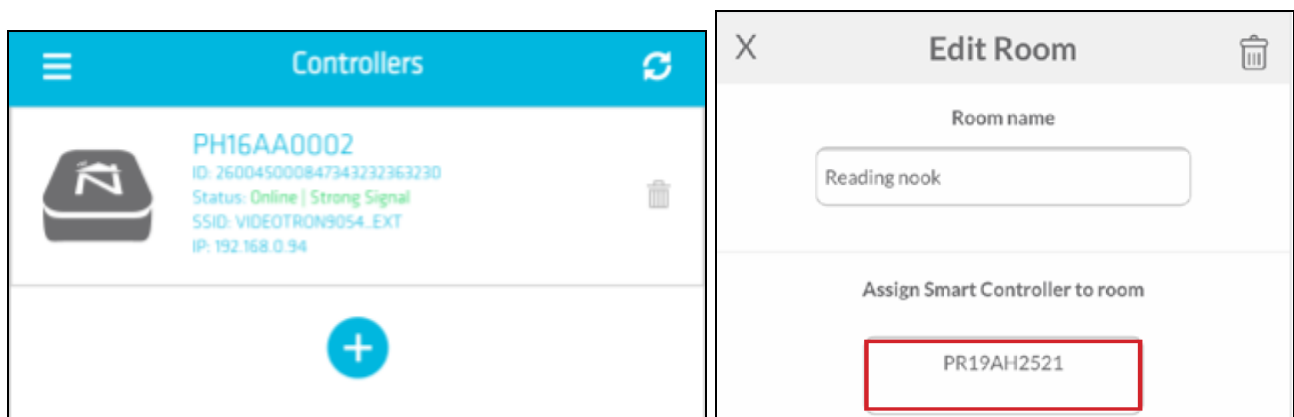
Blinds can be individually or group/room controlled (max 15 per group).

1. Protocol Conventions

Commands are case-insensitive. A carriage return/line feed pair (CRLF) terminates all commands and responses.

Local connection requires previous knowledge of IP and port addresses.

- The controller's local IP can be found using the Neo Smart Blinds app at “Your Controllers.” For accounts with more than one controller, make sure to use the IP address from the controller assigned to the room where the respective blind is located.
- Use port number 8838 (for http/ip communication).
- Using the Neo Smart Blinds app, the controller's channel and IDs can be found on each blinds' advanced page.



Each request/command should open a new connection between the client (app or other devices) and the server (controller). The server will close the connection after every single request/command.



2. Script implementation

Some considerations when implementing this protocol for an automated script.

- The TCP server can only handle a maximum of 1-3 simultaneous client connections. Therefore, it's highly recommended that you only try one connection at a time.
- The server closes each new client connection as soon as the message is received and replied to, which means before the command is validated and finally transmitted. That explains why only an echo message is returned to the client.
- Each received command takes at least 500ms to be verified, processed and transmitted by the controller/hub. [OPTIONAL] Add a delay of at least 100-500ms between each TCP connection and command transmission if many commands must be sent simultaneously.

3. HTTP “transmit” function

3.1. HTTP Request

Syntax: *GET /neo/v1/transmit{?params}*

Params:

<i>command:</i>	"command"	
<i>id:</i>	"controller_id"	
<i>hash:</i>	"unique_hash#"	[optional]

command: - A valid expression containing group_id, channel_id, and action syntax, which is fully described below.

id: - It's the valid controller_id number, composed of 20+ characters. You can find it using the app in “Your Controller”

hash: - It's 7 digits representing an integer number, unique, used to avoid sending more than once a single command when it comes from different sources (e.g. Cloud request). A simple implementation is taking the last 7 digits from time.now() in milliseconds.

Example:

GET

/neo/v1/transmit?command=001.001-00-mu!no&id=380030001047343339383037&hash=2941795



3.2. HTTP Responses:

Syntax: HTTP/1.1 409 Conflict\r\n
 Content-Type: text/html\r\n
 Content-Length: 0\r\n
 Access-Control-Allow-Origin: *\r\n

Response codes	Conditions
200 OK	Message received and transmitted by the Controller
400 Bad Request	Command not found or not valid
401 Unauthorized	Device_ID was not found or not valid
404 Not Found	"/neo/v1/transmit" not found or not valid
409 Conflict	hash # found and format is valid, but # was already used

3.3. Command protocol

length : 16 characters
 syntax : "ID1.ID2-CHANNEL-ACTION!MOTORCODE"

"ID1" : controller byte 1, from integer "000" to "255"
 "." : address separator
 "ID2" : controller byte 2, from integer "000" to "255"
 "_" : channel separator
 "CHANNEL" : channel, individual channel from integer "01" to "14", use channel "15" for a group
 "_" : command separator
 "ACTION" : action, as follows
 "up" : blinds move UP
 "dn" : blinds move DOWN
 "sp" : blinds STOP moving
 "gp" : go to favourite position
 "mu" : micro-step up
 "md" : micro-step down
 "XX" : go to a specific position from 01 to 99. **ONLY** used with motors whose **motor code is in this list "no, db, ra, rb, ap, bl, mb, jo"**.

Options only available for motor code: all 'no' and some 'rx' motors:

"i1" : send blinds to intermediate position 1, replacing command "gp".
 "i2" : send blinds to intermediate position 2



Options are only available when using motor code “tb”, this motor code is always for use with the TDBU (TopDown/BottomUp) blinds or for a room with both “tb” and “bf” motor codes.



Rail selector ref. 1: middle rail, 2: lower rail, 3: both rails.

- "op" : open shades, move both rails UP
- "cl" : close shades, move middle rail UP and lower rail down
- "u2" : move lower rail UP
- "u3" : move both rails UP
- "d2" : move lower rail DOWN
- "d3" : move both rails DOWN
- "o2" : micro-step lower rail UP
- "c2" : micro-step lower rail DOWN
- "gp" : send blinds to the favourite position

Options are only available when using motor codes “mb”, “ra”, “ap”, “db”, AND the motors are of the type TDBU (TopDown/BottomUp), otherwise, use the standard set of actions.



Left for the Lower Rail, Right for the Middle Rail.

- "op" : open shades, move both rails UP
- "cl" : close shades, move middle rail UP and lower rail DOWN
- "u2" : move lower rail UP
- "u4" : move middle rail UP
- "u3" : move both rails UP
- "d2" : move lower rail DOWN
- "d4" : move middle rail DOWN
- "d3" : move both rails DOWN
- "x2" : blinds STOP moving (replaces the “sp” for this type of motor)
- "gp" : send blinds to the favourite position

“!” : exclamation symbol, command separator preceding the motor code

“MOTORCODE”: code designation to indicate the motor protocol to be transmitted

- “bf” : standard Bofu motor
- “vb” : vertical Bofu motor
- “tb” : top-down/bottom-up Bofu motor
- “no” : NEO motor
- “rx” : Raex motor
- “nc” : Nice F-Code motors
- “nr” : Nice 0-Code motors
- “k1” : standard A-OK motor
- “k2” : version 2 A-OK motor (Louvolute exclusive)



“k3” : version 3 A-OK motor (Eclipse exclusive)
“dy” : standard unidirectional Dooya motor
“by” : special code compatible with both Bofu and Dooya motors (**depricated!**)
“wt” : standard Wistar and Alpha motor
“db” : Bi-directional Dooya motor (has optional TDBU type)
“sy” : Somfy RTS motor
“gp” : Gaposia motor
“ra” : Rollease ARC motor (has optional TDBU type)
“re” : Rollease EL motor
“ap” : AMP motors (Turnils) (has optional TDBU type)
“sf” : Sunfrere motor
“rc” : Ronco motor
“bl” : Alta Bliss motor
“mb” : Coulissee MotionBlinds motor (has optional TDBU type)
“jo” : Jiechang motor

Exemples:

- 123.123-05-up!no
- 012.005-15-dn!bf (for controlling a group of blinds)
- 129.021-02-sp!dy
- 129.021-06-87!no

3.4. Finding Controller ID and Channels from APP

Using the app, go to each blind's advanced page to find the necessary ID numbers. See the image below.

All blinds inside the same ROOM have the same unique combination of Controller ID codes. Use CHANNEL = 15 to control all blinds as a group.

To control individual blinds, use the indicated Motor Channel.

That's the same behaviour found using physical remote controls.



Advanced Controls

Please consult your dealer before using this section.

% position ☐

Set Favorite 1

Set Favorite 2

Adj Up Lim Adj Dn Lim

Reverse SFT

Room Code: 242.210-15

Blind Code: 242.210-01

Motor code: bf

Controller Mode: Blind

Blind Code: 242.210-01

Controller ID_1 = 242

Controller ID_2 = 210

Motor Channel = 01

Top-D/Bottom-U: NO

Motor code = bf

Date	version	Change log	Approved by
2024/10/28	1.6	- Initial changelog for the document. - Added commands i1 and i2 for motor codes 'no' and 'rx'.	Antonio
2025/07/03	1.8	- Added a new set of actions for “mb”, “ra”, “ap”, “db” exclusively when they are of the type TDBU. - Added indication to the motor codes with the optional TDBU type.	Antonio